



Statistics Department
Arabic Section
Fourth Level



" Survival Analysis of Breast Cancer Patients Using the Rotterdam Dataset"

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2025-2026

Introduction

Breast cancer is one of the most frequent cancers among women globally and remains a significant public health problem. Breast cancer is characterized by the uncontrolled growth of abnormal cells within the breast that could eventually spread to other places if not treated in time. The mortality rate for breast cancer has declined in recent years because of breakthroughs in screening, diagnosis, and treatment modalities; nevertheless, prognosis remains highly variable between individuals based on different clinical and pathological parameters.

It is important to understand the variables that affect patient survival to develop better treatment methods and predict the outcome of the diseases. In medical studies, examining both the occurrence and the time to occurrence of an event are critical. For patients diagnosed with breast cancer, survival time could be from the time of diagnosis or surgery until death, recurrence, or last contact.

To analyze such a data, the most common approach is through the use of survival analysis. Survival analysis can be defined as a branch of statistics that deals with the estimation of the time to an event of interest. This technique allows the inclusion of observations known as censoring, whereby the event does not occur within the duration of observation.

In this project, survival analysis techniques are applied to the Rotterdam Breast Cancer Dataset which contains clinical information for 2,982 breast cancer patients included in the Rotterdam Tumor Bank. The dataset includes demographic, pathological, treatment, and follow-up variables related to patient survival.

The main objective of this study is to investigate the factors affecting survival among breast cancer patients using Kaplan–Meier survival curves and log-rank tests to compare survival patterns across different patient groups, while the Cox proportional hazards model is applied to evaluate the simultaneous effect of multiple covariates on survival risk.

Descriptive Data Analysis

The Rotterdam breast cancer dataset is obtained from the tumor bank of Rotterdam and was used in Patrick Royston and Douglas Altman (2013), the dataset consists of 2,982 observations and 11 variables, described as follows

Table 1: Rotterdam Breast Cancer Dataset Description

Variable	Description
age	Age at surgery
meno	Menopausal status (0 = premenopausal, 1 = postmenopausal)
size	Tumor size (≤ 20 mm, 20–50 mm, > 50 mm)
grade	Tumor differentiation grade
nodes	Number of positive lymph nodes
pgr	Progesterone receptor level (fmol/L)
er	Estrogen receptor level (fmol/L)
hormon	Hormonal treatment (0 = no, 1 = yes)
chemo	Chemotherapy (0 = no, 1 = yes)
dtime	Time to death or last follow-up (days)
death	Death indicator (0 = alive, 1 = dead)

This dataset contains both **disease-free survival (relapse)** and **overall survival (death)** outcomes, making it suitable for survival analysis techniques such as Kaplan–Meier estimation and Cox regression models.

Table 2: Baseline Characteristics

Characteristic	Overall (N =2982)	Alive (N=1710)	Deceased(N=1272)
Age	55(13)	54(12)	57(14)
Hormonal Treatment			
No Hormonal Therapy	2,643(89%)	1,530(89%)	1,113(88%)
Hormonal Therapy	339(11%)	180(11%)	159(13%)

Tumor Size			
<=20	1,387(47%)	973(57%)	414(33%)
20-50	1,291(43%)	645(38%)	646(51%)
>50	304(10%)	92(5.4%)	212(17%)
Tumor Grade			
2	794(27%)	532(31%)	262(21%)
3	2,188(73%)	1,178(69%)	1,010(79%)
Lymph Nodes			
Low	2,200(74%)	1,464(86%)	736(58%)
High	782(26%)	246(14%)	536(42%)
Menopausal Status			
Premenopausal	1,312 (44%)	844 (49%)	468 (37%)
Postmenopausal	1670(56%)	866 (51%)	804 (63%)
Estrogen Receptors (fmol/l)	167 (272)	158 (255)	178 (294)
Progesterone Receptors (fmol/l)	162 (291)	179 (304)	138 (272)
Chemotherapy			
No Chemotherapy	2,402 (81%)	1,388 (81%)	1,014 (80%)
Chemotherapy	580 (19%)	322 (19%)	258 (20%)

Note: Mean (SD); n (%)

The table shows the baseline characteristics and is stratified by the mortality status by the end of the follow up period (Alive =1710 and Deceased=1272). It provides the frequencies and percentages for categorical variables such as Chemotherapy and the mean and standard deviation for the numerical variables such as Age.

Numerical variables:

Figure 1 shows the distribution of the age at the time of surgery. The figure illustrate that the distribution is bell shaped and symmetric and the highest frequency is approximately between 45 and 65.

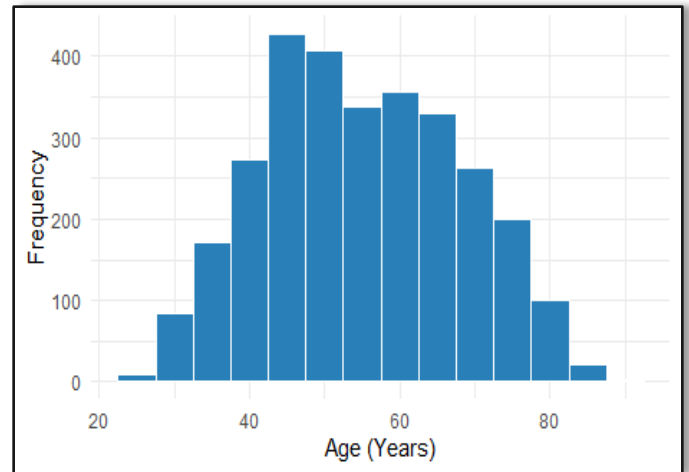


Figure 1: Age Distribution at Surgery

Figure 2 provides a histogram of Estrogen Receptor which is clear that its right skewed distribution. The majority of the cohort are very close to zero (ER) while a long tail extends towards values exceeding 1000 (ER). This skewness justifies the summary in **Table 2** and why the (SD) was a large number and exceeds the mean.

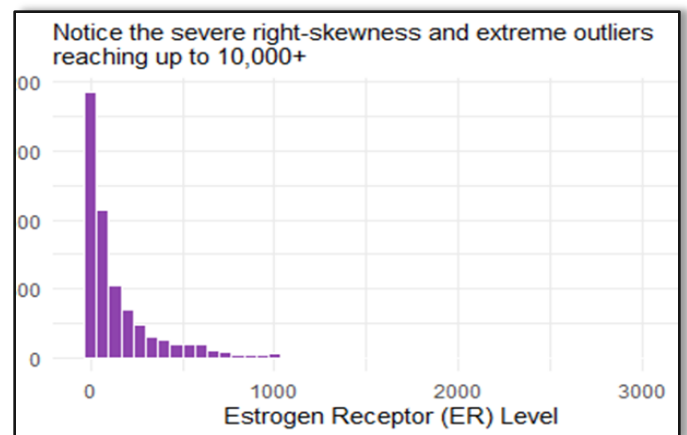


Figure 2: Estrogen Receptor

Figure 3 provides the same conclusion that we get in the last paragraph that the Progesterone Receptor (PGR) is a right skewed distribution with majority of patients very close to zero (PGR) and a long tail extends towards values exceeding 1000 (PGR).

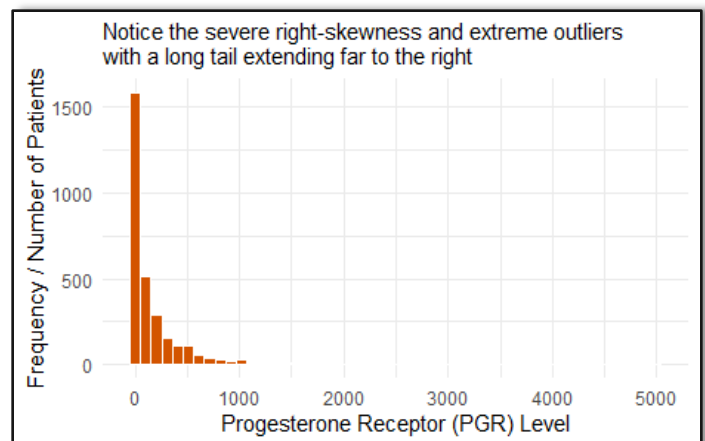


Figure 3: Progesterone

Categorical Variables:

Figure 4 shows the overall survival status of the cohort at the final follow up. Out of 2982 patient, 1710 remained alive (censored) while 1272 patient were deceased.

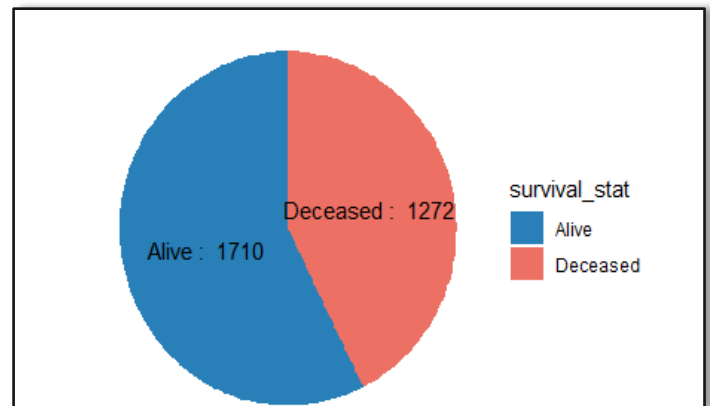


Figure 4: Mortality Status

Figure 5 shows the distribution of hormonal treatment. It is highly imbalanced, and the majority is for the patients with no hormonal therapy.

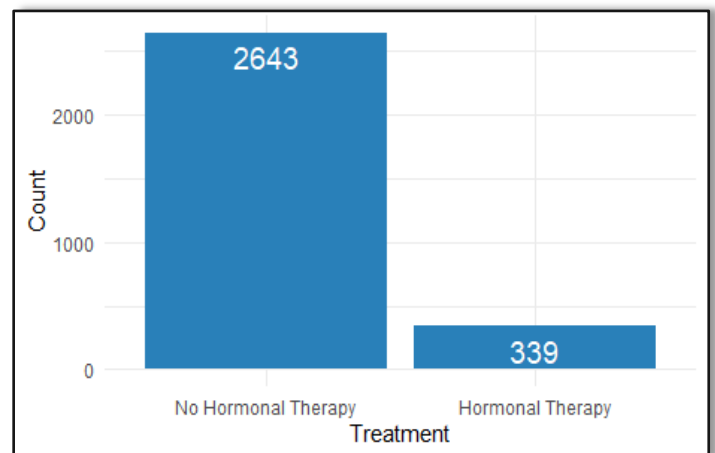


Figure 5: Hormonal Treatment Usage

Figure 6 shows the distribution of baseline tumor size. The small sizes are the most prevalent, with 1387 patients with tumors ≤ 20 mm and 1291 patients with tumors between 20-50 mm, while the patients with tumor > 50 mm represent the smallest category with only 304 patients.

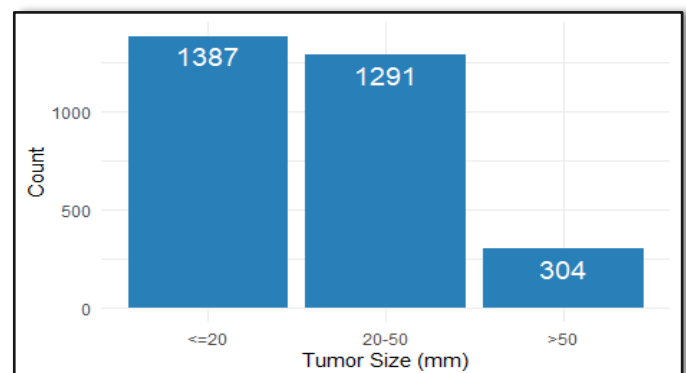


Figure 6: Tumor Size Distribution

Figure 7 indicates that the majority of the cohort suffer from aggressive disease as the patients with tumors of grade 3 represent around 73%.

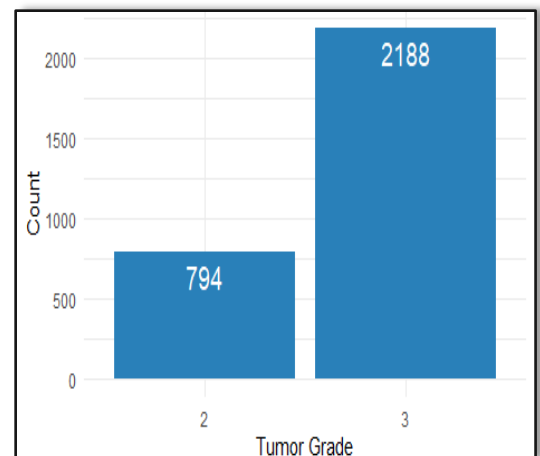


Figure 7: Tumor Grade Distribution

Figure 8 describe the lymph node involvement categorized into two groups. 2200 patients had low nodal involvement while 25% of cohort presented with high nodal involvement.

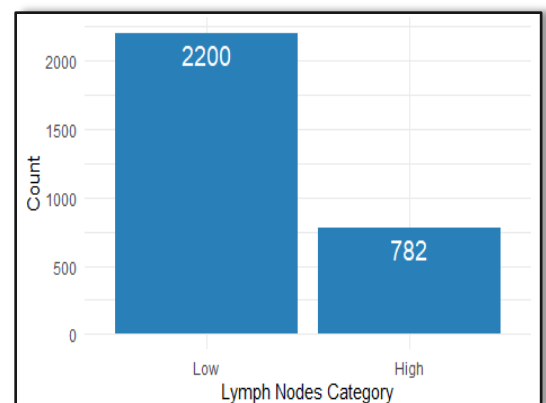


Figure 8: Lymph Nodes

Figure 9 shows the distribution of Chemotherapy. The data reveals a highly imbalanced distribution where the around 81% of the cohort didn't receive a chemotherapy.

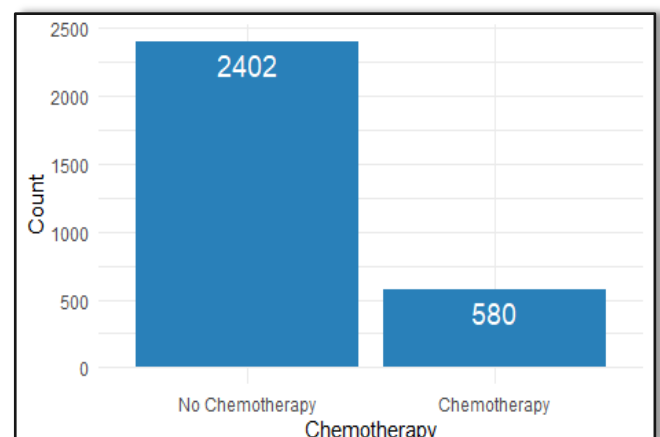


Figure 9: Chemotherapy distribution

Survival Analysis

In this study, we measure the time from a clearly defined **start point** “Primary Breast Cancer Surgery” to a clearly defined **end point** “Death due to Breast Cancer” while accounting for several clinical variables that may influence this duration, such as hormonal treatment, progesterone receptor status, estrogen receptor status, among others.

The Kaplan–Meier estimator was used in this study cause unlike the Life Table method, Kaplan–Meier calculates survival probabilities at each exact event time, making it more accurate for clinical datasets with varying follow-up times.

The majority of patients (57.3%) were censored, meaning more than half of the cohort did not experience the event during the study period. This is typical and expected in survival studies it confirms that survival analysis is the appropriate method, as it correctly handles these censored observations rather than excluding them.

A substantial proportion (42.7%) experienced the event, which is high enough to provide reliable survival estimates and gives the analysis adequate statistical power to detect differences between groups.

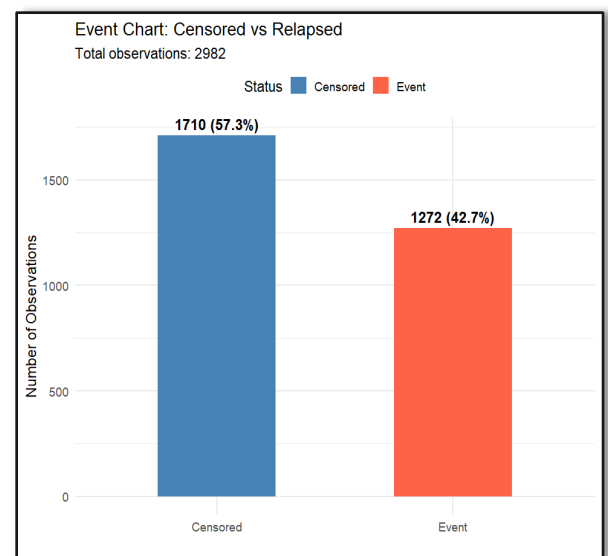


Figure 12: Event Chart (Censored vs Relapse)

The Overall Kaplan-Meier Survival Curve:

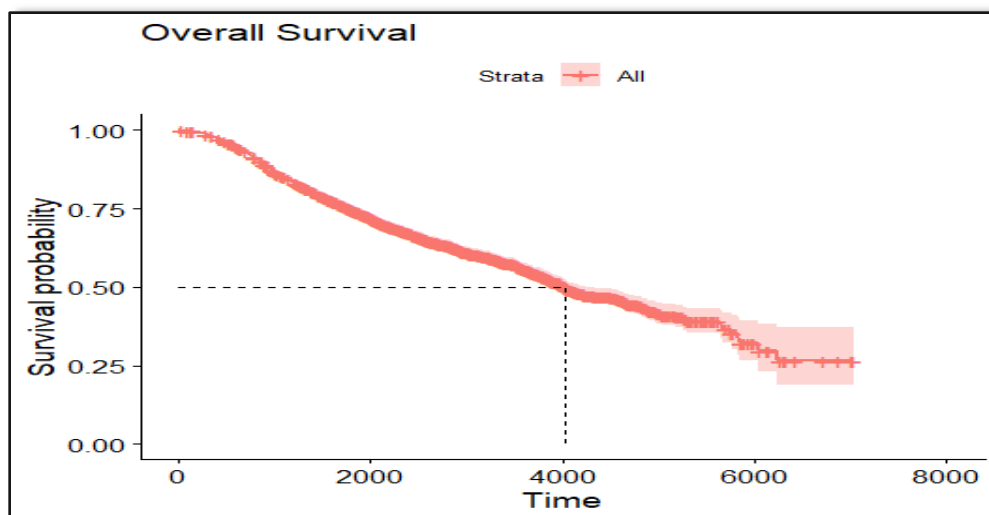


Figure 13: Kaplan-Meier Survival Function Estimate

Table 3: Overall Kaplan–Meier Survival Estimates

Median	0.95LCL	0.95UCL
4033	3888	4309

Based on both the figure and the table, starting from the day of surgery, all 2,982 patients were alive, giving a survival probability of 1.00 (100%). From that point, the curve gradually steps downward over time, reflecting the fact that patients were dying throughout the follow-up period and the assumption that $s(t)$ decreases over time.

The most important number from the table is the median survival of 4,033 days, which is roughly 11 years after surgery. What this means that half of the patients had died by the 11-year mark, and the other half were still alive beyond it.

As for how confident we are in that estimate, the 95% confidence interval runs from 3,888 to 4,309 days, which is roughly a 10.6-to-11.6-year range.

Things get a bit less certain toward the end of the follow-up, beyond around 16 years (6,000 days), where the red band becomes very wide and the curve starts making bigger and it is becoming constant.

Kaplan-Meier Survival Curves by Key Variables:

1. Hormon Variable:

From early in the follow-up, the two curves separate, with the untreated group showing consistently higher survival probability throughout the entire follow-up period. The median survival was approximately 4,020 days (11 years) for the untreated group versus 2,800 days (7.7 years) for the treated group, meaning that untreated patients survived on average 3.3 years longer before reaching the 50% survival mark.

And the log-rank test confirmed a statistically significant difference between the two groups

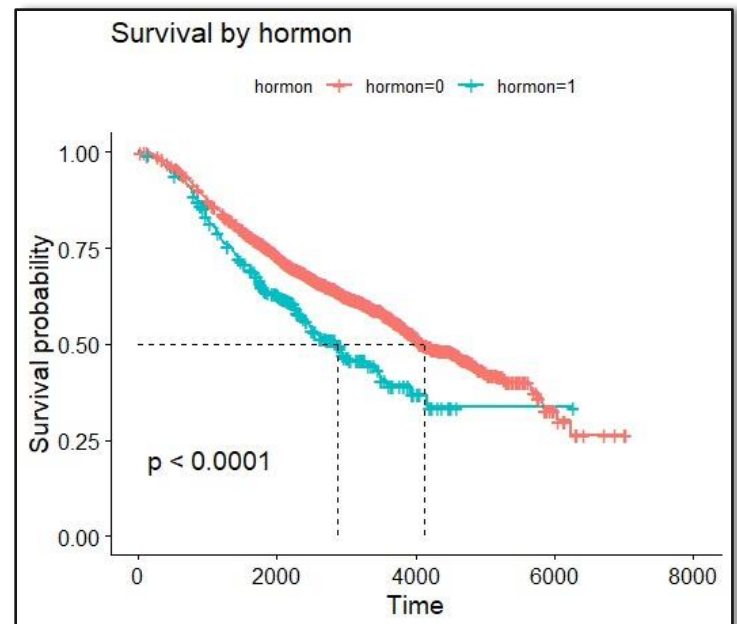


Figure 14: Kaplan-Meier Survival Curve by Hormon Variable

($p < 0.0001$), and the proportional hazards assumption was verified using cox.zph test ($p = 0.72$), indicating that the hazard ratio between the two groups remains constant over time, and therefore the log-rank result is considered valid and reliable.

2. Size Variable:

The three curves separate clearly and early, showing a consistent pattern the larger the tumor, the worse the survival throughout the entire follow-up period. The median survival was approximately 5,500 days (15 years) for the smallest size group ($\leq 20\text{mm}$), 3,300 days (9 years) for the intermediate group (20–50mm), and 1980 days (5.5 years) for the largest size group ($> 50\text{mm}$), meaning that patients with the smallest tumors survived on average 9.5 years longer before reaching the 50% survival mark compared to those with the largest tumors.

The log-rank test confirmed a statistically significant difference between the three groups ($p < 0.0001$). However, the proportional hazards assumption was violated as indicated by the cox.zph test ($p = 0.011$) then the hazard ratio between the size groups does not remain constant over time, meaning the log-rank result should be interpreted with caution.

3. Grade Variable:

From early in the follow-up, the two curves separate, with Grade 2 patients showing consistently higher survival probability throughout the entire follow-up period. The median survival was approximately 5,700 days (15.6 years) for Grade 2 versus 3,500 days (9.6 years) for Grade 3, meaning that Grade 2 patients survived on average 6 years longer before

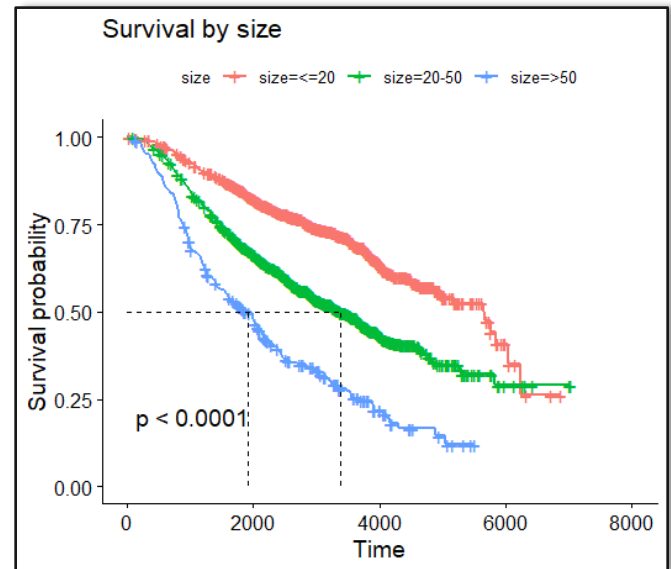


Figure 15: Kaplan-Meier Survival Curve by Size Variable

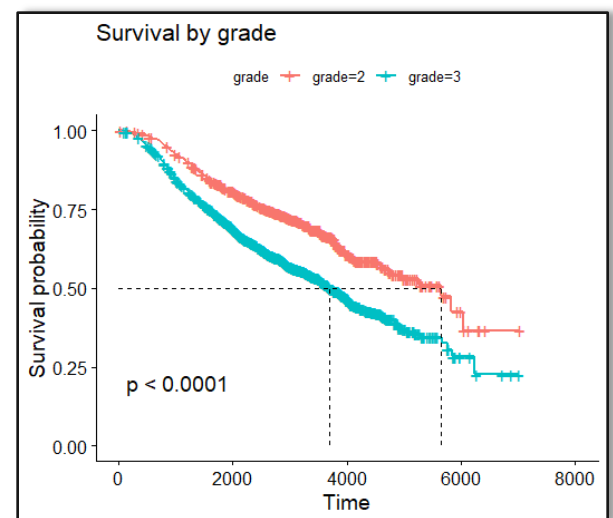


Figure 16: Kaplan-Meier Survival Curve by Grade Variable

reaching the 50% survival mark compared to Grade 3 patients. The log-rank test confirmed a statistically significant difference between the two groups ($p < 0.0001$), and the proportional hazards assumption was verified using the cox.zph test ($p = 0.06$) confirming that the log-rank result is valid and reliable.

4.Nodes Variable:

From early in the follow-up, the two curves separate rapidly and widely more than any other variable examined with the Low node group showing consistently higher survival probability throughout the entire follow-up period. The median survival was approximately 5,000 days (13.7 years) for the Low group versus 2,000 days (5.5 years) for the High group, meaning that patients with low node involvement survived on average 8.2 years longer before reaching the 50% survival mark compared to those with high nodal involvement.

The log-rank test confirmed a statistically significant difference between the two groups ($p < 0.0001$) and the proportional hazards assumption was violated as indicated by the cox.zph test ($p = 0.00023$) and therefore the log-rank result should be interpreted with caution.

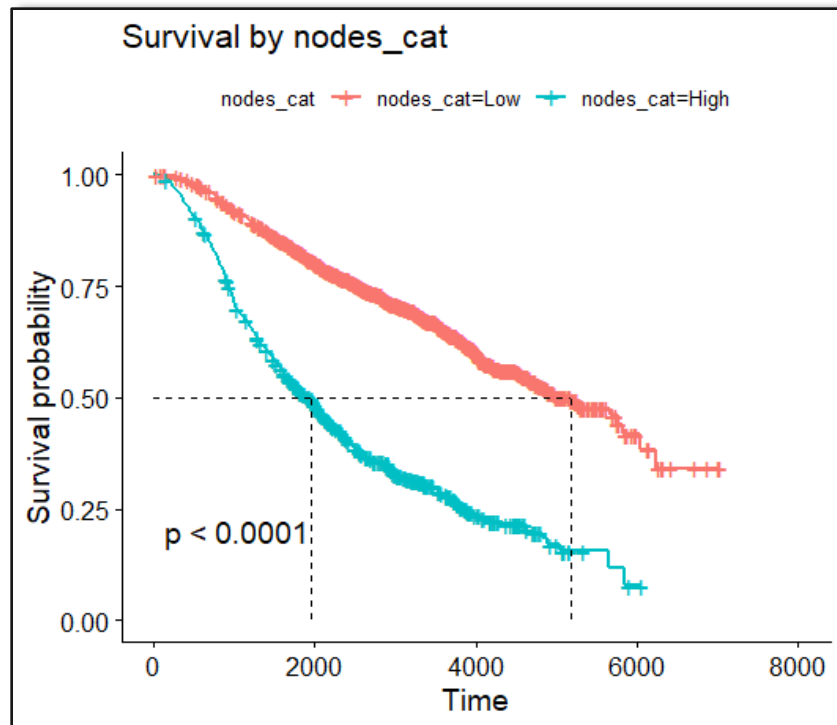


Figure 17: Kaplan-Meier Survival Curve by Nodes Variable

Cox proportional hazards regression

Cox regression was used to evaluate the association between clinical and pathological variables and survival among breast cancer patients. A full multivariable Cox proportional hazards model was first fitted including hormonal treatment group, age, menopausal status, number of positive lymph nodes, tumor size, tumor grade, estrogen receptor level, progesterone receptor level, and chemotherapy status. The model was used to estimate hazard ratios and identify factors associated with mortality risk over time.

```
> summary(cox_full)
Call:
coxph(formula = Surv(ttr, relapse) ~ grp + age + meno + nodes +
      size + grade + er + pgr + chemo, data = rotterdam)

n= 2982, number of events= 1272

              coef exp(coef)    se(coef)      z Pr(>|z|)
grp1      -6.553e-02  9.366e-01  8.840e-02 -0.741 0.458535
age        1.406e-02  1.014e+00  3.830e-03  3.671 0.000242 ***
menoPost   7.046e-02  1.073e+00  1.006e-01  0.701 0.483583
nodes      7.295e-02  1.076e+00  4.879e-03 14.953 < 2e-16 ***
size20-50  4.425e-01  1.557e+00  6.536e-02  6.771 1.28e-11 ***
size>50    8.222e-01  2.276e+00  9.142e-02  8.993 < 2e-16 ***
grade3     3.156e-01  1.371e+00  7.082e-02  4.456 8.33e-06 ***
er         -5.512e-05  9.999e-01  1.107e-04 -0.498 0.618466
pgr        -3.676e-04  9.996e-01  1.226e-04 -2.998 0.002720 **
chemoYes    5.032e-02  1.052e+00  8.198e-02  0.614 0.539342
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The full model showed that survival was mainly influenced by age, lymph node involvement, tumor size, tumor grade, and progesterone receptor level, where higher values of the clinical risk factors were associated with worse survival, while progesterone receptor showed a protective effect. Other variables, including hormonal treatment, menopausal status, estrogen receptor, and chemotherapy, were not statistically significant after adjustment.

Based on these results, a reduced model was then fitted including only the significant predictors to obtain a more parsimonious and interpretable model while maintaining predictive performance.

The Reduced Model:

$$RR(t|x) = \exp (0.01482*Age + 0.07333*Nodes + 0.43982*Size20-50 + 0.81959*Size>50 + 0.31516*Grade3 - 0.000384*PGR)$$

Table 4: Results of the Cox Proportional Hazards Reduced Model

Variable	Hazard Ratio (HR)	95% CI Lower	95% CI Upper	p-value
age	1.0149	1.0105	1.0194	3.55e-11 ***
nodes	1.0761	1.0661	1.0862	9.63e-54 ***
size20–50	1.5524	1.3662	1.7641	1.53e-11 ***
size>50	2.2696	1.9005	2.7103	1.40e-19 ***
grade3	1.3705	1.1933	1.5740	8.15e-06 ***
pgr	0.9996	0.9994	0.9998	0.00109 **

Likelihood ratio test= 523 on 6 df, p=<2e-16

Concordance= 0.692 (se = 0.008)

The overall Cox proportional hazards model was highly statistically significant, indicating that the included covariates jointly explain variation in survival time. The model also demonstrated good discriminatory ability, meaning it can reasonably distinguish between patients with higher and lower risk of death.

- **Age (HR = 1.0149):** Each additional year of age is associated with a 1.5% increase in the hazard of death, holding other variables constant.
- **Number of positive lymph nodes (HR = 1.0761):** Each additional positive lymph node is associated with a 7.6% increase in the hazard of death, holding other variables constant.
- **Tumor size (20–50 vs ≤20) (HR = 1.55):** Patients with tumors sized 20–50 have a 55% higher hazard of death compared to those with tumors ≤20, holding other variables constant.
- **Tumor size (>50 vs ≤20) (HR = 2.27):** Patients with tumors larger than 50 have more than double the hazard of death compared to those with tumors ≤20, holding other variables constant.
- **Grade3 (HR = 1.37):** Patients with Grade 3 tumors have a 37% higher hazard of death compared to Grade 2, holding other variables constant.
- **Progesterone receptor (HR = 0.9996):** Higher progesterone receptor levels are associated with a slight decrease in the hazard of death, holding other variables constant.

Forest Plot of Hazard Ratios (HRs):

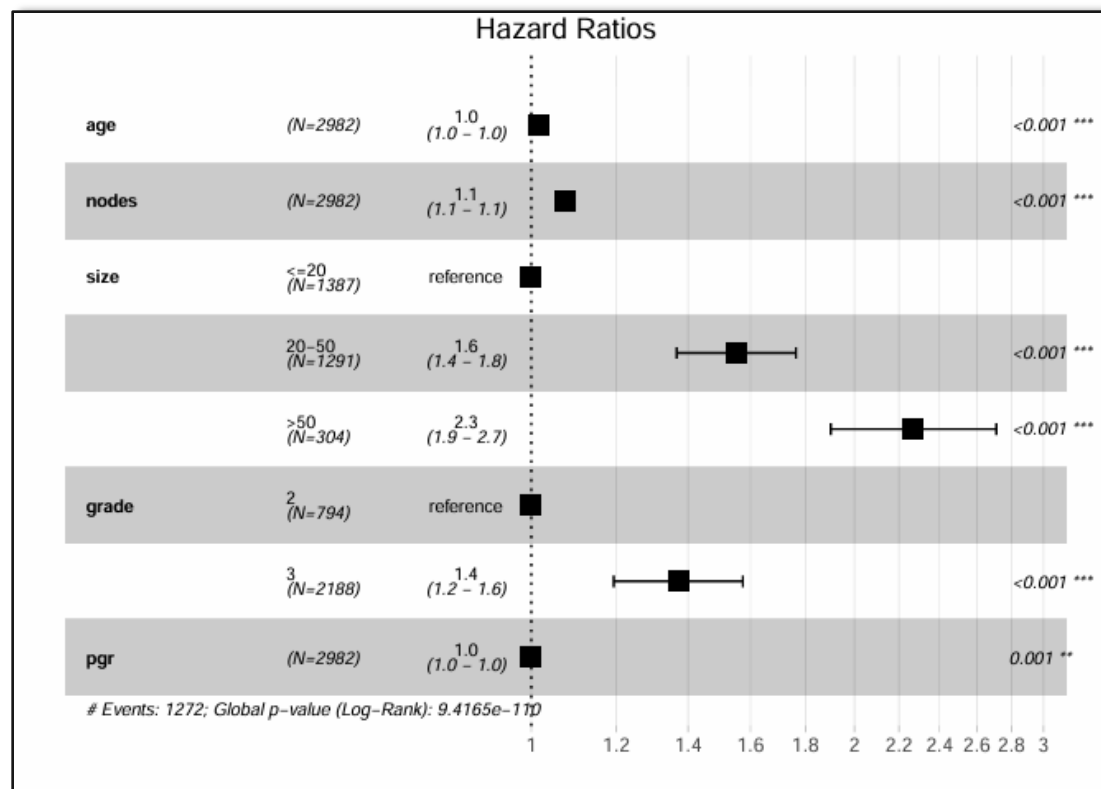


Figure 18: Forest Plot of Hazard Ratios from the Cox Proportional

The figure presents the results of a Cox proportional hazards model, showing hazard ratios (HRs) with 95% confidence intervals for each predictor. The model is highly statistically significant overall (global log-rank p-value $\approx 9.4 \times 10^{-110}$), indicating strong evidence that the included variables jointly influence survival. Age shows a statistically significant but very small effect, with a hazard ratio close to 1, suggesting a minimal increase in risk with increasing age. The number of positive lymph nodes is associated with an increased hazard, with each additional node raising the risk of death by approximately 10%. Tumor size demonstrates a clear dose-response relationship, where patients with tumors measuring 20–50 have about a 60% higher hazard compared to those with tumors ≤ 20 , while those with tumors > 50 have more than double the risk. Tumor grade also shows an important effect, as Grade 3 tumors are associated with approximately a 40% higher hazard compared to Grade 2. Progesterone receptor level has a hazard ratio close to 1, indicating a very small effect on survival despite statistical significance. Overall, the results suggest that tumor size and lymph node involvement are the most influential predictors of survival in this model.

Checking the Proportional Hazards Assumption:

The proportional hazards assumption in the Cox proportional hazards model was assessed using Schoenfeld residuals. This test evaluates whether the effect of each covariate on the hazard remains constant over time.

The results indicated that the assumption was violated for several variables. Age showed a statistically significant violation, indicating that its effect on the hazard is not constant over time. Similarly, the progesterone receptor variable (pgr) strongly violated the assumption, suggesting clear time-dependent effects. The number of lymph nodes was borderline significant, indicating a possible mild violation. In contrast, tumor size and grade did not show statistically significant evidence of violation, suggesting that the proportional hazards assumption is reasonably satisfied for these variables. Overall, the global test was highly significant, indicating that the proportional hazards assumption is violated for the model as a whole. These findings suggest that some covariates may have time-varying effects and should be further investigated, potentially using stratification or time-dependent covariates.

	chisq	df	p
age	14.44	1	0.00014
nodes	3.86	1	0.04933
size	5.13	2	0.07677
grade	3.07	1	0.07970
pgr	41.54	1	1.2e-10
GLOBAL	59.39	6	6.0e-11

Addressing Proportional Hazards Violations

The initial assessment of the PH assumption through Schoenfeld residuals revealed significant violations for age, number of lymph nodes, and progesterone receptor (pgr) levels. These results indicated that the hazard ratios for these specific variables were not constant over time, rendering the standard Cox model estimates potentially biased.

The Extended Cox Model:

To resolve these violations, an Extended Cox proportional hazards model was fitted. Instead of assuming constant hazards, I incorporated time-dependent covariates using a time-transform function, $tt(x, t) = x \times \log(t + 1)$.

This approach allows the impact of age, nodes, and pgr to vary as follow-up time increases, providing a mathematically valid representation of their effects over the entire study period.

Final Model Results:

```
> summary(cox_time_dep)
Call:
coxph(formula = Surv(ttr, relapse) ~ tt(age) + tt(nodes) + size +
      grade + tt(pgr), data = rotterdam, tt = function(x, t, ...) x *
      log(t + 1))

n= 2982, number of events= 1272

              coef exp(coef)    se(coef)      z Pr(>|z|)
tt(age)      2.089e-03  1.002e+00  3.088e-04  6.766 1.32e-11 ***
tt(nodes)    9.988e-03  1.010e+00  6.617e-04 15.094 < 2e-16 ***
size20-50    4.440e-01  1.559e+00  6.523e-02  6.806 1.00e-11 ***
size>50      8.363e-01  2.308e+00  9.027e-02  9.265 < 2e-16 ***
grade3       3.193e-01  1.376e+00  7.068e-02  4.518 6.25e-06 ***
tt(pgr)     -4.222e-05  1.000e+00  1.553e-05 -2.720 0.00654 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The overall Cox proportional hazards model with time-dependent covariates was highly statistically significant, indicating that the included variables jointly explain the variation in survival time. The model demonstrated good discriminatory ability with a concordance of **0.691**.

- **Age (tt(age)):** When accounting for time-dependent effects, each additional year of age is associated with a 0.2% increase in the hazard of relapse over time, holding other variables constant.
- **Number of positive lymph nodes (tt(nodes)):** The effect of lymph node involvement is time-dependent; for each additional positive node, the hazard of relapse increases by 1% as follow-up progresses, holding other variables constant.
- **Tumor size (20–50 vs ≤20):** Patients with tumors sized 20–50 mm have a 56% higher hazard of relapse compared to those with tumors ≤20 mm, holding other variables constant.
- **Tumor size (>50 vs ≤20):** Patients with tumors larger than 50 mm have 2.31 times the hazard of relapse compared to those with tumors ≤20 mm, holding other variables constant.

- **Tumor Grade:** Patients with Grade 3 tumors have a 38% higher hazard of relapse compared to those with Grade 2 tumors, holding other variables constant.
- **Progesterone receptor (tt(pgr)):** While statistically significant ($p = 0.0065$), the progesterone receptor level shows a very small time-dependent effect on the hazard of relapse.

By utilizing the tt() function, the model now accounts for the non-constant effects of age, lymph nodes, and progesterone receptors identified in the Schoenfeld residual analysis. This adjustment ensures that the model provides a more accurate representation of mortality risk over time compared to the initial standard Cox model. Consistent with previous findings, tumor size and lymph node involvement remain the most influential predictors of survival in this patient cohort.

Conclusion:

In this study, survival analysis techniques were applied to the Rotterdam Breast Cancer dataset to examine the factors influencing patient survival. The analysis demonstrated that several clinical and pathological variables, including age, tumor size, lymph node involvement, and disease stage, play an important role in determining survival outcomes. Kaplan–Meier survival curves were useful in visualizing differences between patient groups, while the log-rank test helped assess the statistical significance of these differences.

Furthermore, the Cox proportional hazards model provided a deeper understanding of how multiple variables simultaneously affect the risk of death, allowing for the identification of the most significant predictors of survival. These methods together offered a comprehensive view of survival patterns within the dataset.

Overall, this study emphasizes the value of survival analysis in medical research, as it not only helps in understanding the progression of diseases like breast cancer but also supports clinicians in making informed decisions regarding diagnosis, prognosis, and treatment planning.